Hardness Testing Lab

(Lab - 8)

What is hardness ?

- Resistance to localized plastic (permanent) deformation of a material.
- It can be used for many engineering applications to achieve the basic requirement of mechanical property.

Hardness Measurement Techniques

- **Scratch Method**

Scratch resistance of various materials through the ability of a harder material to scratch a softer material.

- **Rebound or dynamic hardness**

Measures the height of the "bounce" of a diamondtipped hammer dropped from a fixed height onto a material.

- **Indentation Methods**

A shaped tool is driven into the surface of a specimen by applying known forces. The size and area of the indentation is used to calculate a **hardness value** for the material.

Scratch Method (Mohs Scale)

Rebound or dynamic hardness

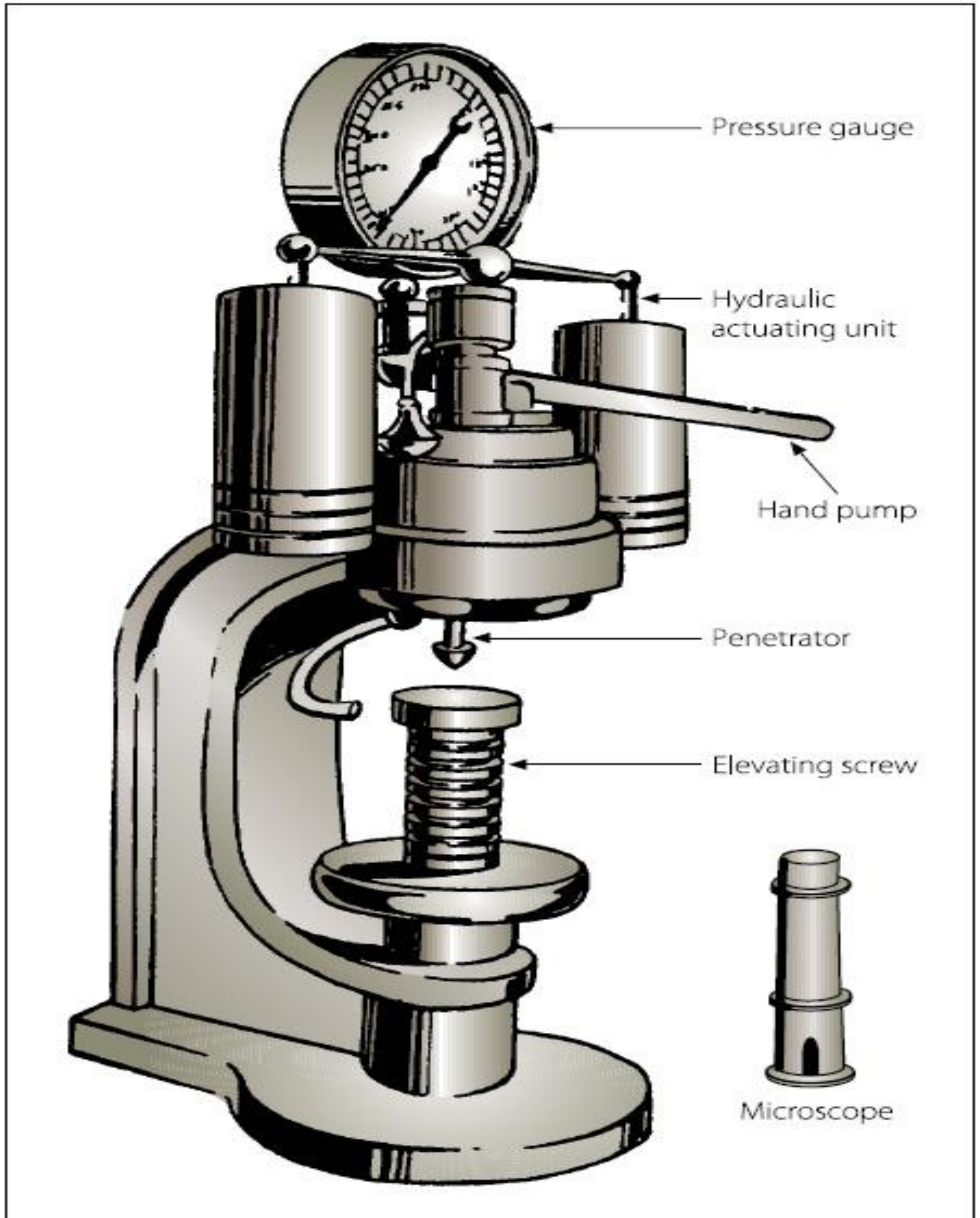
- measures hardness of materials from the height of a rebound indenter' dropped on the materials surface with
- The device is known as the 'Scleroscope'.

Indentation Methods

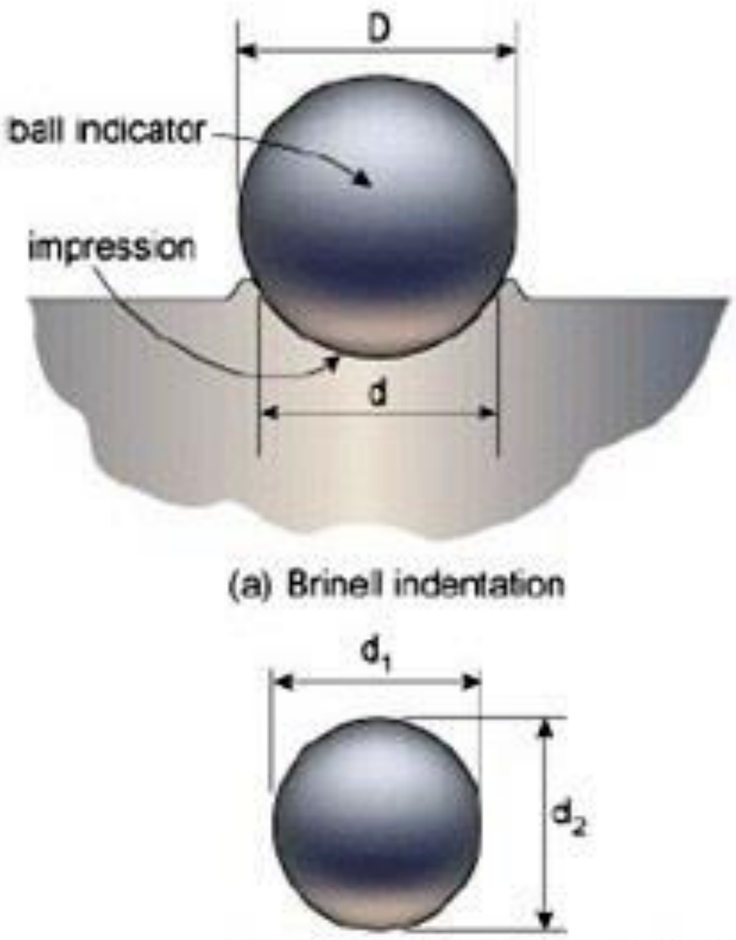
- Brinell Hardness
- Vickers hardness
- Rockwell hardness
- Nano indentation

Brinell Hardness

- 10mm diameter hardened steel or tungsten carbide ball
- Load range of 500-3000 kg, depending on hardness of the particular material
- The load is applied for a standard time for 30 second



Brinell Hardness



(a) Brinell indentation

(b) measurement of impression diameter

$$BHN = \frac{P}{\frac{\pi D}{2}\left[D - \sqrt{D^2 - d^2}\right]}$$

Where:

P is the test load [kg]

D is the diameter of the ball [mm]

d is the average impression diameter of indentation [mm]

Brinell advantages and disadvantages

Advantages:

- Can use different loads to test both soft and hard materials.
- Not easily affected by surface scratches or roughness.
- Good for testing materials with uneven or mixed internal structure.

Disadvantages:

- Not suitable for small parts, because the big indentation might cause cracks or weak spots.

Vickers Hardness

- Vickers hardness test uses a square-base diamond pyramid
- Angle between opposite faces of the pyramid of 136˚
- The Vickers test can be used for all metals and has one of the widest scales among hardness tests

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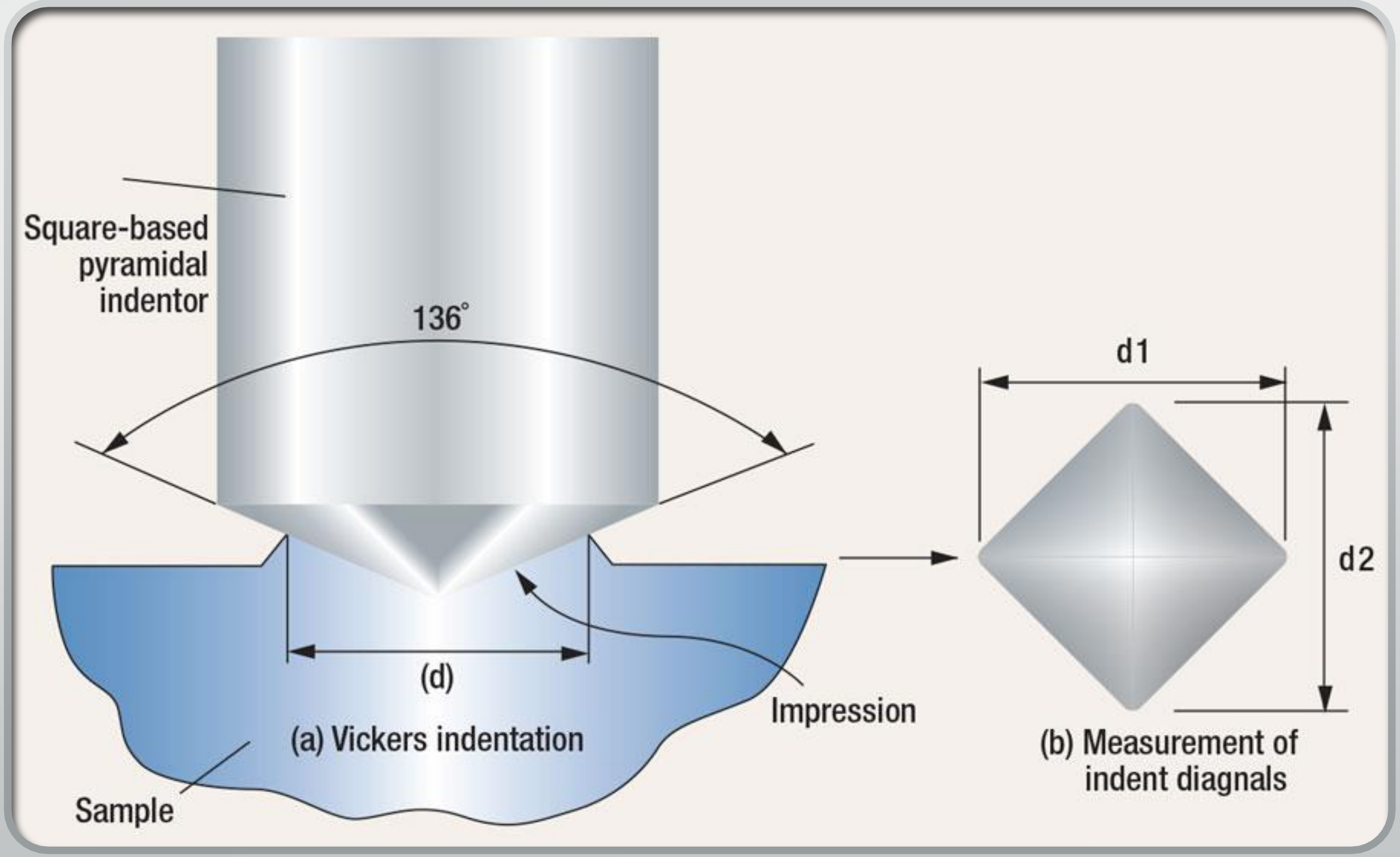
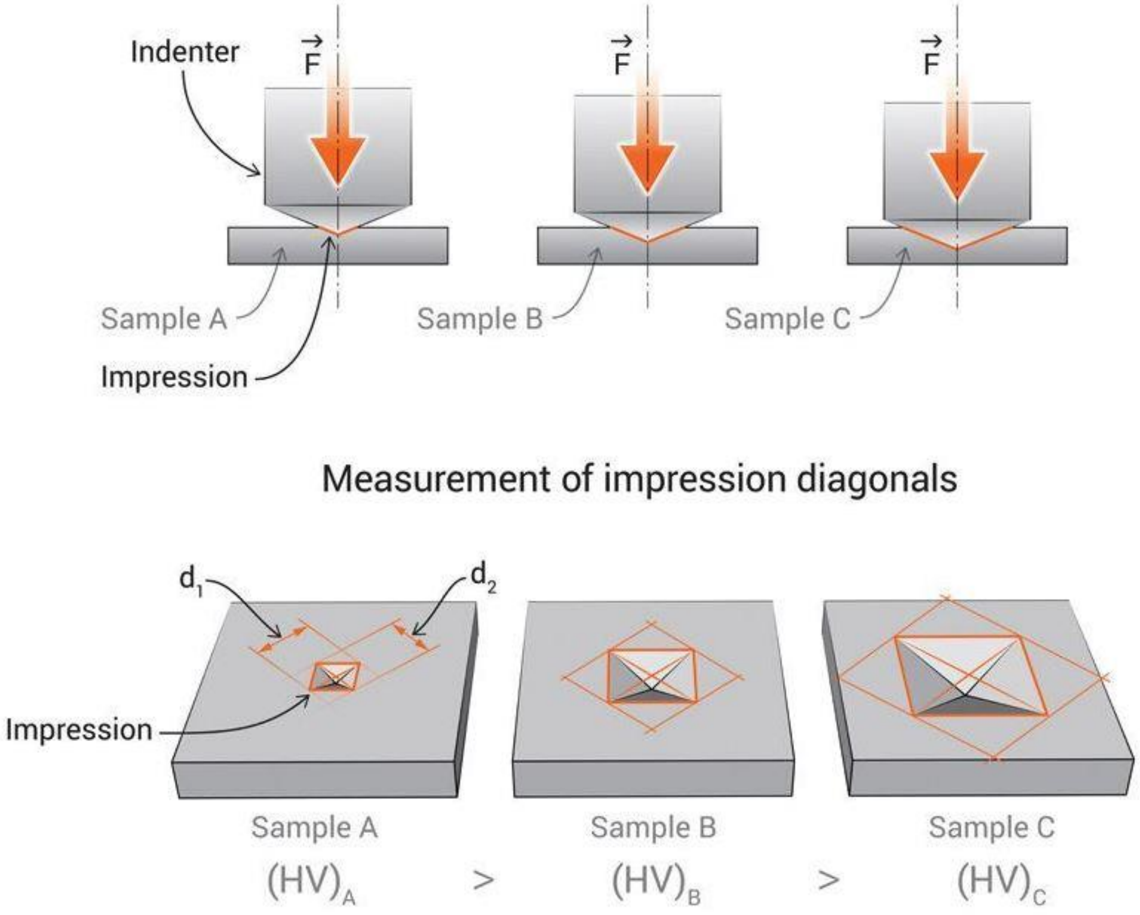



Indenter
$\vec{F}$
$\vec{F}$
$\vec{F}$
Sample A
Sample B
Sample C
Impression
Measurement of impression diagonals
d_1
d_2
Impression
Sample A
Sample B
Sample C
$(HV)_A > (HV)_B > (HV)_C$

Vickers Hardness Testing

- Loads ranging from 1-120 kg
- Applied for between 10 and 15 seconds
- Where **F** is the applied load, kg
- **D** is the average length of diagonals, mm

$$H_V = \frac{2F\sin\frac{\theta}{2}}{D^2}$$ Where θ=136°

$$H_V = \frac{1.8544F}{D^2}$$

Rockwell Hardness

- The Rockwell Hardness test is a hardness measurement based on the net increase in depth of impression as a constant load is applied
- Minor Load = 10 kg
- Major Load = varies
- There are different scales of Rockwell hardness. Which depends on the indenter specification and load

Working of the Rockwell Test

- Place the specimen close to the indenter.
- Applied the minor load and a zero reference position is established
- The major load is applied for a specified time period (dwell time).
- The major load is released leaving the minor load applied.
- Take reading from the analog dial of the Rockwell hardness machine.

ROCKWELL HARDNESS SCALE

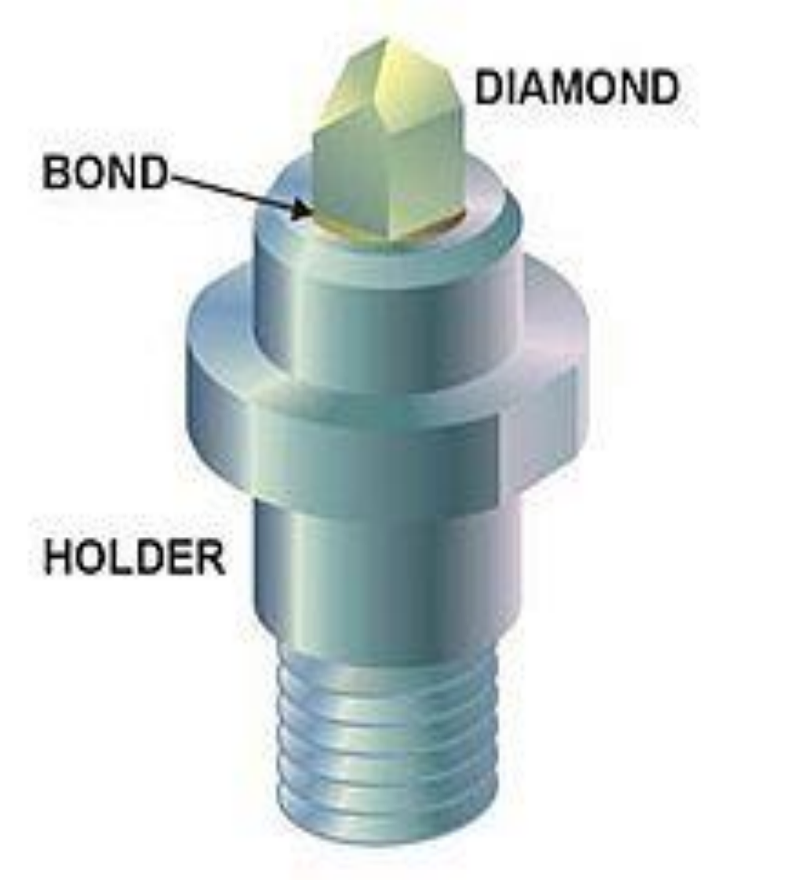
Scale	Major Load	Indenter shape	Indenter Material
A	60	Cone	Diamond
B	100	Sphere Ball 1/16 Diameter	Steel
C	150	Cone	Diamond

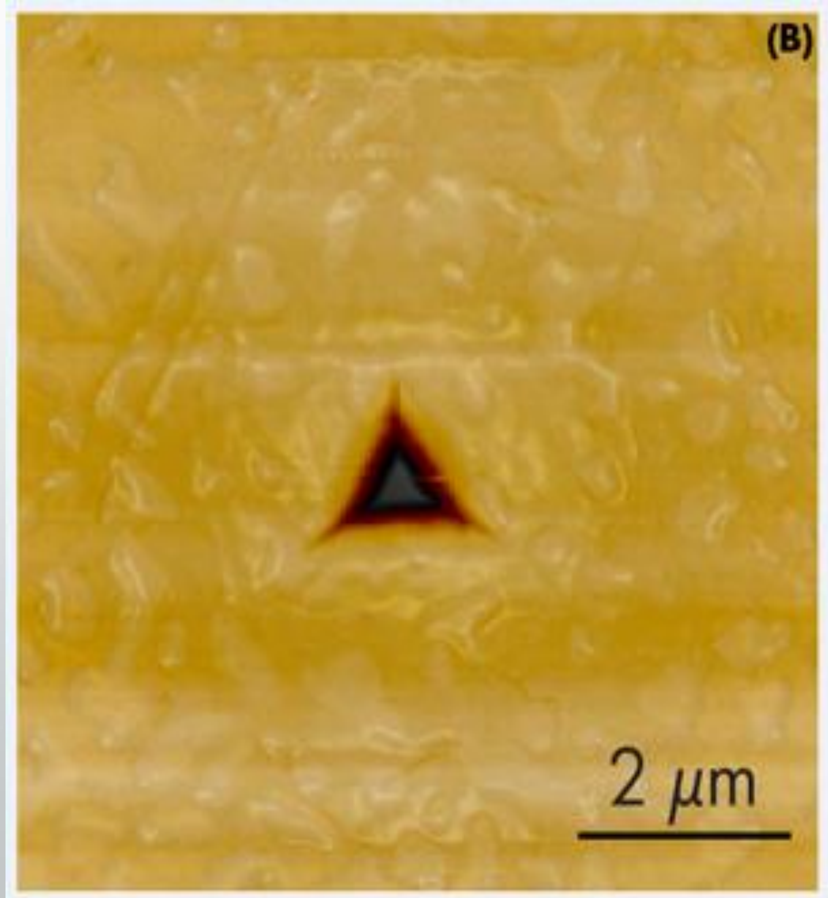
Advantages of Rockwell Test

- Its speed
- Freedom from personal error.
- Ability to distinguish small hardness difference
- Small size of indentation

Nano Indentation

- Used to measure the hardness of hard thin film coatings
- The indenter does not penetrate the substrate





Parameter	Brinell Hardness (HB)	Vickers Hardness (HV)	Rockwell Hardness (HR)	Nanoindentation
Minimum Load	500 N	1 N	10 N	<1 µN
Maximum Load	3000 N	120 N	1500 N	Up to a few mN
Materials Tested	Metals, castings, forgings	Metals, thin films, coatings	Metals, polymers, ceramics	Thin films, biological materials, nanostructures
Accuracy	Moderate	High	High	Very high (nanoscale resolution)
Advantages	Good for coarse materials	Suitable for very small parts	Quick and easy to use	High precision and repeatability
	Large indentation eliminates	Universal across material types	Minimal surface preparation	Provides detailed mechanical properties
	local material variations			(e.g., elastic modulus, hardness)
Disadvantages	Limited to larger samples	Time consuming	Limited accuracy for soft materials	Expensive equipment
	Surface preparation required	Requires careful alignment	Cannot measure very small areas	Requires expert handling
Unit of Measurement	HB	HV	HR	GPa (for modulus, hardness)

Thank you